

CST413	MACHINE LEARNING	CATEGORY	L	T	P	CREDIT	YEAR OF INTRODUCTION
		PEC	2	1	0	3	2019

Preamble: This course enables the learners to understand the advanced concepts and algorithms in machine learning. The course covers the standard and most popular supervised learning algorithms such as linear regression, logistic regression, decision trees, Bayesian learning and the Naive Bayes algorithm, basic clustering algorithms and classifier performance measures. This course helps the students to provide machine learning based solutions to real world problems.

Prerequisite: Basic understanding of probability theory, linear algebra and Python Programming

Course Outcomes: After the completion of the course the student will be able to

CO1	Illustrate Machine Learning concepts and basic parameter estimation methods. (Cognitive Knowledge Level: Apply)
CO2	Demonstrate supervised learning concepts (regression, linear classification). (Cognitive Knowledge Level: Apply)
CO3	Illustrate the concepts of Multilayer neural network and Support Vector Machine (Cognitive Knowledge Level: Apply)
CO4	Describe unsupervised learning concepts and dimensionality reduction techniques. (Cognitive Knowledge Level: Apply)
CO5	Solve real life problems using appropriate machine learning models and evaluate the performance measures (Cognitive Knowledge Level: Apply)

Mapping of course outcomes with program outcomes

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												
CO2												
CO3												
CO4												
CO5												

Abstract POs defined by National Board of Accreditation			
PO#	Broad PO	PO#	Broad PO
PO1	Engineering Knowledge	PO7	Environment and Sustainability
PO2	Problem Analysis	PO8	Ethics
PO3	Design/Development of solutions	PO9	Individual and team work
PO4	Conduct investigations of complex problems	PO10	Communication
PO5	Modern tool usage	PO11	Project Management and Finance
PO6	The Engineer and Society	PO12	Life long learning

Assessment Pattern

Bloom's Category	Continuous Assessment Tests		End Semester Examination Marks (%)
	Test 1 (%)	Test 2 (%)	
Remember	30	30	30
Understand	30	30	30
Apply	40	40	40
Analyze			
Evaluate			
Create			

Mark Distribution

Total Marks	CIE Marks	ESE Marks	ESE Duration
150	50	100	3

Continuous Internal Evaluation Pattern:

Attendance	10 marks
Continuous Assessment Tests(Average of Internal Tests 1 & 2)	25 marks
Continuous Assessment Assignment	15 marks

Internal Examination Pattern

Each of the two internal examinations has to be conducted out of 50 marks. First series test shall be preferably conducted after completing the first half of the syllabus and the second series test shall be preferably conducted after completing remaining part of the syllabus. There will be two parts: Part A and Part B. Part A contains 5 questions (preferably, 2 questions each from the completed modules and 1 question from the partly completed module), having 3 marks for each question adding up to 15 marks for part A. Students should answer all questions from Part A. Part B contains 7 questions (preferably, 3 questions each from the completed modules and 1 question from the partly completed module), each with 7 marks. Out of the 7 questions, a student should answer any 5.

End Semester Examination Pattern:

There will be two parts; Part A and Part B. Part A contains 10 questions with 2 questions from each module, having 3 marks for each question. Students should answer all questions. Part B contains 2 full questions from each module of which student should answer any one. Each question can have maximum 2 sub-divisions and carries 14 marks.

Syllabus**Module-1 (Overview of machine learning)**

Machine learning paradigms-supervised, semi-supervised, unsupervised, reinforcement learning. Basics of parameter estimation - maximum likelihood estimation(MLE) and maximum a posteriori estimation(MAP). Introduction to Bayesian formulation.

Module-2 (Supervised Learning)

Regression - Linear regression with one variable, Linear regression with multiple variables, solution using gradient descent algorithm and matrix method, basic idea of overfitting in regression. Linear Methods for Classification- Logistic regression, Naive Bayes, Decision tree algorithm ID3.

Module-3 (Neural Networks (NN) and Support Vector Machines (SVM))

Perceptron, Neural Network - Multilayer feed forward network, Activation functions (Sigmoid, ReLU, Tanh), Backpropagation algorithm.

SVM - Introduction, Maximum Margin Classification, Mathematics behind Maximum Margin Classification, Maximum Margin linear separators, soft margin SVM classifier, non-linear SVM, Kernels for learning non-linear functions, polynomial kernel, Radial Basis Function(RBF).

Module-4 (Unsupervised Learning)

Clustering - Similarity measures, Hierarchical Agglomerative Clustering, K-means partitional clustering, Expectation maximization (EM) for soft clustering. Dimensionality reduction – Principal Component Analysis.

Module-5 (Classification Assessment)

Classification Performance measures - Precision, Recall, Accuracy, F-Measure, Receiver Operating Characteristic Curve(ROC), Area Under Curve(AUC). Bootstrapping, Cross Validation, Ensemble methods, Bias-Variance decomposition. Case Study: Develop a classifier for face detection.

Text Book

1. Ethem Alpaydin, Introduction to Machine Learning, 2nd edition, MIT Press 2010.
2. Mohammed J. Zaki and Wagner Meira, Data Mining and Analysis: Fundamental Concepts and Algorithms, Cambridge University Press, First South Asia edition, 2016.
3. Jake VanderPlas, Python Data Science Handbook, O'Reilly Media, 2016
4. Tom Mitchell, Machine Learning, McGraw-Hill, 1997.

Reference Books

1. Christopher Bishop. Neural Networks for Pattern Recognition, Oxford University Press, 1995.
2. Kevin P. Murphy. Machine Learning: A Probabilistic Perspective, MIT Press 2012.
3. Trevor Hastie, Robert Tibshirani, Jerome Friedman, The Elements Of Statistical Learning, Second edition Springer 2007.
4. P. Langley, Elements of Machine Learning, Morgan Kaufmann, 1995.
5. Richert and Coelho, Building Machine Learning Systems with Python.
6. Davy Cielen, Arno DB Meysman and Mohamed Ali. Introducing Data Science: Big Data, Machine Learning, and More, Using Python Tools, Dreamtech Press 2016.

Course Level Assessment Questions

Course Outcome1 (CO1):

1. A coin is tossed 100 times and lands heads 62 times. What is the maximum likelihood estimate for θ , the probability of heads.
2. Suppose data $x_1, \dots, x_n$ are independent and identically distributed drawn from an exponential distribution $\exp(\lambda)$. Find the maximum likelihood for λ .
3. Suppose $x_1, \dots, x_n$ are independent and identically distributed(iid) samples from a distribution with density

Find the maximum estimate(MLE) for θ .

$$f_X(x|\theta) = \begin{cases} \frac{\theta x^{\theta-1}}{3^\theta}, & 0 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases} \quad \text{likelihood}$$

4. Find the maximum likelihood estimator (MLE) and maximum a posteriori (MAP) estimator for the mean of a univariate normal distribution. Assume that we have N samples, $x_1, \dots, x_N$ independently drawn from a normal distribution with known variance σ^2 and unknown mean μ and the prior distribution for the mean is itself a normal distribution with mean ν and variance β^2 . What happens to the MLE and MAP estimators as the number of samples goes to infinity.

Course Outcome 2(CO2):

1. Suppose that you are asked to perform linear regression to learn the function that outputs y , given the D -dimensional input x . You are given N independent data points, and that all the D attributes are linearly independent. Assuming that D is around 100, would you prefer the closed form solution or gradient descent to estimate the regressor?
2. Suppose you have a three class problem where class label $y \in 0, 1, 2$ and each training example X has 3 binary attributes $X_1, X_2, X_3 \in 0, 1$. How many parameters (probability distribution) do you need to know to classify an example using the Naive Bayes classifier?

Course Outcome 3(CO3):

1. What are support vectors and list any three properties of the support vector classifier solution?
2. Why do you use kernels to model a projection from attributes into a feature space, instead of simply projecting the dataset directly?

- Describe how Support Vector Machines can be extended to make use of kernels. Illustrate with reference to the Gaussian kernel $K(x, y) = e^{-z}$, where $z = (x-y)^2$.
- Briefly explain one way in which using tanh instead of logistic activations makes optimization easier.
- ReLU activation functions are most used in neural networks instead of the tanh activation function. Draw both activation functions and give a) an advantage of the ReLU function compared to the tanh function. b) a disadvantage of the ReLU function compared to the tanh function.

Course Outcome 4(CO4): .

- Which similarity measure could be used to compare feature vectors of two images? Justify your answer.
- Illustrate the strength and weakness of k-means algorithm.
- Suppose you want to cluster the eight points shown below using **k**-means

	A_1	A_2
x_1	2	10
x_2	2	5
x_3	8	4
x_4	5	8
x_5	7	5
x_6	6	4
x_7	1	2
x_8	4	9

Assume that $k = 3$ and that initially the points are assigned to clusters as follows:

$C_1 = \{x_1, x_2, x_3\}$, $C_2 = \{x_4, x_5, x_6\}$, $C_3 = \{x_7, x_8\}$. Apply the **k**-means algorithm until convergence, using the Manhattan distance.

- Cluster the following eight points representing locations into three clusters: $A_1(2, 10)$, $A_2(2, 5)$, $A_3(8, 4)$, $A_4(5, 8)$, $A_5(7, 5)$, $A_6(6, 4)$, $A_7(1, 2)$, $A_8(4, 9)$.

Initial cluster centers are: $A_1(2, 10)$, $A_4(5, 8)$ and $A_7(1, 2)$.

The distance function between two points $a = (x_1, y_1)$ and $b = (x_2, y_2)$ is defined as $D(a, b) = |x_2 - x_1| + |y_2 - y_1|$

Use **k**-Means Algorithm to find the three cluster centers after the second iteration.

Course Outcome 5(CO5):

1. What is ensemble learning? Can ensemble learning using linear classifiers learn classification of linearly non-separable sets?
2. Describe boosting. What is the relation between boosting and ensemble learning?
3. Classifier A attains 100% accuracy on the training set and 70% accuracy on the test set. Classifier B attains 70% accuracy on the training set and 75% accuracy on the test set. Which one is a better classifier. Justify your answer.
4. What are ROC space and ROC curve in machine learning? In ROC space, which points correspond to perfect prediction, always positive prediction and always negative prediction? Why?
5. Suppose there are three classifiers A,B and C. The (FPR, TPR) measures of the three classifiers are as follows – A (0, 1), B (1, 1) , C (1,0.5). Which can be considered as a perfect classifier? Justify your answer.

Model Question Paper**QP CODE:****Reg No:** _____**Name:** _____**PAGES : 4****APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY****SEVENTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR****Course Code: CST413****Course Name: Machine Learning****Max. Marks : 100****Duration: 3 Hours****PART A****Answer All Questions. Each Question Carries 3 Marks**

1. Define supervised learning? Name special cases of supervised learning depending on whether the inputs/outputs are categorical, or continuous.

2. Differentiate between Maximum Likelihood estimation (MLE) and Maximum a Posteriori (MAP) estimation?
3. What is overfitting and why is it a problem?
4. Specify the basic principle of gradient descent algorithm.
5. Suppose that you have a linear support vector machine(SVM) binary classifier. Consider a point that is currently classified correctly, and is far away from the decision boundary. If you remove the point from the training set, and re-train the classifier, will the decision boundary change or stay the same? Justify your answer.
6. Mention the primary motivation for using the kernel trick in machine learning algorithms?
7. Expectation maximization (EM) is designed to find a maximum likelihood setting of the parameters of a model when some of the data is missing. Does the algorithm converge? If so, do you obtain a locally or globally optimal set of parameters?
8. Illustrate the strength and weakness of k-means algorithm.
9. Classifier A attains 100% accuracy on the training set and 70% accuracy on the test set. Classifier B attains 70% accuracy on the training set and 75% accuracy on the test set. Which one is a better classifier. Justify your answer.
10. How does bias and variance trade-off affect machine learning algorithms?

(10x3=30)

Part B

(Answer any one question from each module. Each question carries 14 Marks)

11. (a) Suppose that X is a discrete random variable with the following probability mass function: where $0 \leq \theta \leq 1$ is a parameter. The following 10 independent observations (7)

X	0	1	2	3
$P(X)$	$2\theta/3$	$\theta/3$	$2(1 - \theta)/3$	$(1 - \theta)/3$

were taken from such a distribution: (3, 0, 2, 1, 3, 2, 1, 0, 2, 1). What is the maximum likelihood estimate of θ .

- (b) Suppose you have a three class problem where class label $y \in \{0, 1, 2\}$ (7)
and each training example X has 3 binary attributes $X_1, X_2, X_3 \in \{0, 1\}$.
How many parameters (probability distribution) do you need to know

to classify an example using the Naive Bayes classifier?

OR

12. (a) Consider the geometric distribution, which has p.m.f $P(X = k) = (1 - \theta)^{k-1} \theta$. Assume that n i.i.d data are drawn from that distribution. (7)

- i. Write an expression for the log-likelihood of the data as a function of the parameter θ .
- ii. Find the maximum likelihood estimate for θ ?
- ii. Let θ has a beta prior distribution. What is the posterior distribution of θ ?

- (b) Find the maximum a posteriori (MAP) estimator for the mean of a univariate normal distribution. Assume that we have N samples, $\mathbf{x}_1, \dots, \mathbf{x}_N$ independently drawn from a normal distribution with known variance σ^2 and unknown mean μ and the prior distribution for the mean is itself a normal distribution with mean ν and variance β^2 . (7)

13. (a) Consider the hypothesis for the linear regression $h_\theta(\mathbf{x}) = \theta_0 + \theta_1 \mathbf{x}$, and the cost function $J(\theta_0, \theta_1) = 1/2m \sum_i^m (h_\theta(\mathbf{x}^{(i)}) - y^{(i)})^2$ where m is the number of training examples. Given the following set of training examples. (7)

$\mathbf{x}$	$\mathbf{y}$
3	2
1	2
0	1
4	3

Answer the following questions :

- 1) Find the value of $h_\theta(2)$ if $\theta_0 = 0$ and $\theta_1 = 1.5$
- 2) Find the value of $J(0,1)$
- 3) Suppose the value of $J(\theta_0, \theta_1) = 0$. What can be inferred from this.

- (b) Assume we have a classification problem involving 3 classes: professors, students, and staff members. There are 750 students, 150 staff members and 100 professors. All professors have blond hair, 50 staff members have blond hair, and 250 students have blond hair. Compute the information gain of the test "hair color = blond" that returns true or false. (3)

- (c) Explain the significance of regularization. How do Ridge differs from Lasso regularization? (4)

OR

14. (a) The following dataset can be used to train a classifier that determines whether a given person is likely to own a car or not. There are three features: education level (primary, secondary, or university); residence (city or country); gender (female, male). (7)

education	residence	gender	has car?
sec	country	female	yes
univ	country	female	yes
prim	city	male	no
univ	city	male	no
sec	city	female	no
sec	country	male	yes
prim	country	female	yes
univ	country	male	yes
sec	city	male	yes
prim	city	female	no
univ	city	female	no
prim	country	male	yes

Use ID3 Algorithm and find the best attribute at the root level of the tree

- (b) Consider a linear regression problem $y = w_1x + w_0$, with a training set having m examples $(x_1, y_1), \dots, (x_m, y_m)$. Suppose that we wish to minimize the mean 5th degree error (loss function) given by $1/m \sum_1^m (y_i - w_1x_i - w_0)^5$. (7)
1. Calculate the gradient with respect to the parameter w_1 .
 2. Write down pseudo-code for on-line gradient descent on w_1 .
 3. Give one reason in favor of on-line gradient descent compared to batch-gradient descent, and one reason in favor of batch over on-line.

15. (a) Consider a support vector machine whose input space is 2-D, and the inner products are computed by means of the kernel $K(\mathbf{x}, \mathbf{y}) = (\mathbf{x} \cdot \mathbf{y} + 1)^2 - 1$, where $\mathbf{x} \cdot \mathbf{y}$ denotes the ordinary inner product. Show that the mapping to feature space that is implicitly defined by this kernel is the mapping to 5-D given by (8)

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \rightarrow \phi(\mathbf{x}) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2} x_1 x_2 \\ \sqrt{2} x_1 \\ \sqrt{2} x_2 \end{bmatrix}$$

- (b) Consider a neuron with four inputs, and weight of edge connecting the inputs are 1, 2, 3 and 4. Let the bias of the node is zero and inputs are 2, 3, 1, 4. If the activation function is linear $f(x)=2x$, compute the output of the neuron. (3)
- (c) Compare ReLU with Sigmoid function (3)

OR

16. (a) State the mathematical formulation to express Soft Margin as a constraint optimization problem. (10)

- (b) What is the basic idea of back propagation algorithm (4)

17. (a) Suppose that we have the following data (one variable). Use single linkage Agglomerative clustering to identify the clusters. (8)

Data: (2, 5, 9, 15, 16, 18, 25, 33, 33, 45).

- (b) Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8): (6)
- (i) Compute the Euclidean distance between the two objects.
 - (ii) Compute the Manhattan distance between the two objects.
 - (iii) Compute the Minkowski distance between the two objects, using $p = 3$

OR

18. (a) Suppose that we have the following data: (8)

(2, 0), (1, 2), (2, 2), (3, 2), (2, 3), (3, 3), (2, 4), (3, 4), (4, 4), (3, 5)

Identify the cluster by applying the k-means algorithm, with $k = 2$. Try using initial cluster centers as far apart as possible

(b) Describe EM algorithm for Gaussian Mixtures (8)

19. (a) Suppose the dataset had 9700 cancer-free images from 10000 images from cancer patients. Find precision, recall and accuracy ? Is it a good classifier? Justify. (7)

Actual Class\Predicted class	cancer = yes	cancer = no	Total
cancer = yes	90	210	300
cancer = no	140	9560	9700
Total	230	9770	10000

(b) What is Principal Component Analysis (PCA)? Which eigen value indicates the direction of largest variance? (7)

OR

20. (a) Assume you have a model with a high bias and a low variance. What are the characteristics of such a model? (6)

(b) What are ROC space and ROC curve in machine learning? In ROC space, which points correspond to perfect prediction, always positive prediction and always negative prediction? Why? (8)

Teaching Plan

No	Contents	No. of Lecture Hours (37 hrs)
Module -1 (Overview of machine learning) (7 hours)		
1.1	Supervised, semi-supervised, unsupervised learning, reinforcement learning (Text Book (TB) 1: Chapter 1)	1 hour
1.2	Maximum likelihood estimation(MLE) (TB 1: Section 4.2)	1 hour
1.3	Maximum likelihood estimation (MLE)- example (TB 1: Section 4.2)	1 hour
1.4	Maximum a posteriori estimation(MAP) (TB 4: Section 6.2)	1 hour
1.5	Maximum a posteriori estimation(MAP)-example (TB 4: Section 6.2)	1 hour
1.6	Bayesian formulation (TB 1: Section 14.1, 14.2)	1 hour
1.7	Bayesian formulation -example (TB 1: Section 14.1, 14.2)	1 hour
Module-2 (Supervised Learning) (7 hours)		
2.1	Linear regression with one variable (TB 1: Section 2.6)	1 hour
2.2	Multiple variables, Solution using gradient descent algorithm and matrix method (No derivation required) (TB 1: Section 5.8)	1 hour
2.3	Overfitting in regression, Lasso and Ridge regularization	1 hour
2.4	Logistic regression	1 hour
2.5	Naive Bayes (TB 2: Section 18.2)	1 hour
2.6	Decision trees (TB 2: Chapter 19)	1 hour
2.7	Decision trees- ID3 algorithm (TB 2: Chapter 19)	1 hour
Module-3 (Neural Networks and Support Vector Machines) (9 hours)		
3.1	Perceptron, Perceptron Learning	1 hour
3.2	Multilayer Feed forward Network, Activation Functions (Sigmoid, ReLU, Tanh)	1 hour
3.3	Back Propagation Algorithm	1 hour
3.4	Illustrative Example for Back Propagation	1 hour
3.5	Introduction, Maximum Margin Hyperplane,	1 hour
3.6	Mathematics behind Maximum Margin Classification	1 hour
3.7	Formulation of maximum margin hyperplane and solution	1 hour

3.8	Soft margin SVM, Solution of Soft margin SVM	1 hour
3.9	Non-linear SVM , Kernels for learning non-linear functions, Examples - Linear, RBF, Polynomial	1 hour
Module-4 (Unsupervised Learning) (7 hours)		
4.1	Similarity measures- Minkowski distance measures(Manhattan, Euclidean), Cosine Similarity	1 hour
4.2	Clustering - Hierarchical Clustering (TB 2: Chapter 14)	1 hour
4.3	K-means partitional clustering (TB 2: Chapter 13)	1 hour
4.4	Expectation maximization (EM) for soft clustering (TB 2: Chapter 13)	1 hour
4.5	Expectation maximization (EM) for soft clustering (TB 2: Chapter 13)	1 hour
4.6	Dimensionality reduction – Principal Component Analysis (TB 1: Section 6.3)	1 hour
4.7	Dimensionality reduction – Principal Component Analysis (TB 1: Section 6.3)	1 hour
Module-5 (Classification Assessment) (7 hours)		
5.1	Performance measures - Precision, Recall, Accuracy, F-Measure, ROC, AUC. (TB 2: Chapter 22.1)	1 hour
5.2	Boot strapping, Cross validation	1 hour
5.3	Ensemble methods- bagging, boosting	1 hour
5.4	Bias-Variance decomposition (TB 2: Chapter 22.3)	1 hour
5.5	Bias-Variance decomposition (TB 2: Chapter 22.3)	1 hour
5.6	Face detection (TB 3: Chapter 5 Section Application: A Face Detection Pipeline)	1 hour
5.7	Face detection (TB 3: Chapter 5 Section Application: A Face Detection Pipeline)	1 hour